

Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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Background

- **MRCTs are the global standard**; pooling decisions hinge on effect modifier similarity. **ICH E17** describes pooling qualitatively — **no quantitative metric or threshold**.
- **Existing measures fall short**: SMD detects only location ($N(50, 5^2)$ vs. $N(50, 15^2)$ gives SMD = 0); KS lacks clinical scale; KL is asymmetric and tail-unstable.
- **Goal**. A dissimilarity index capturing location, scale, and shape on a clinically interpretable scale.

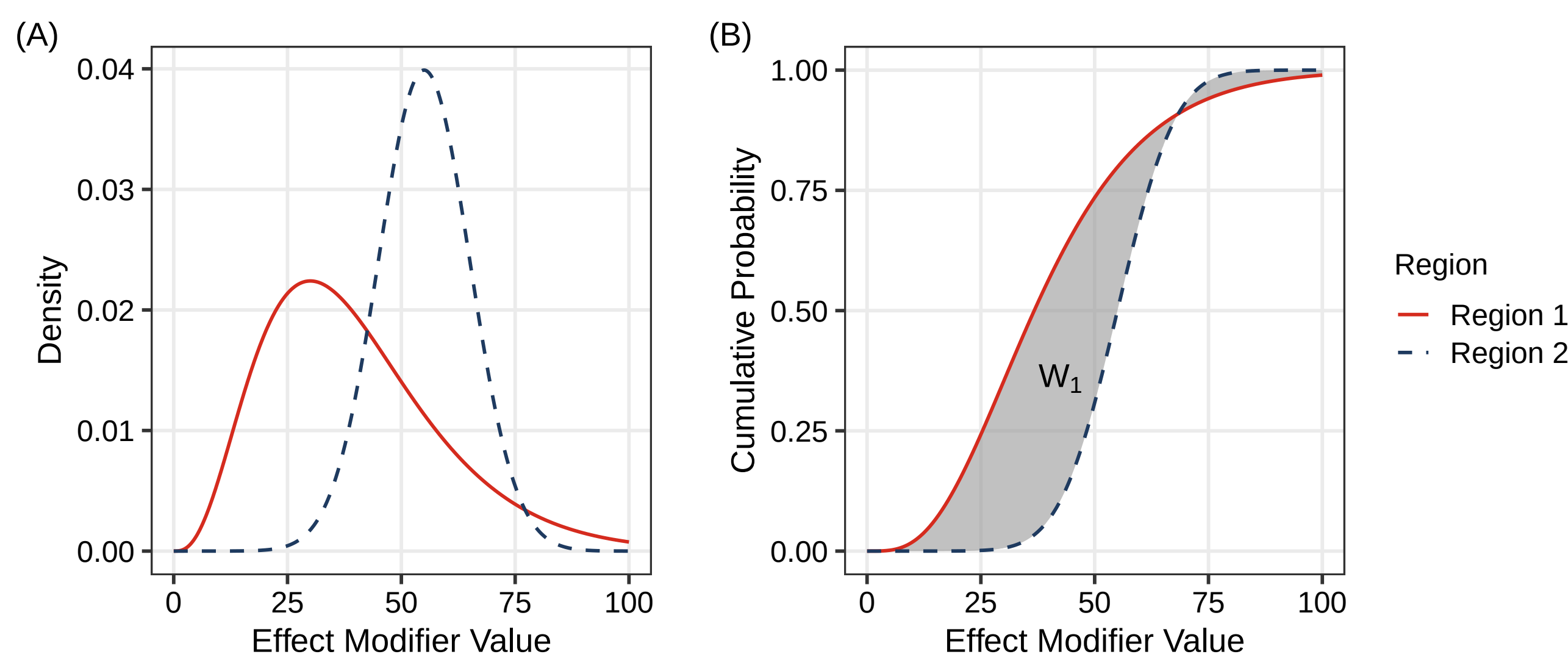
Methods

Definition. The normalized Area Between CDFs Distance is the Wasserstein-1 distance between region CDFs F_1 and F_2 , normalized by a robust pooled scale:

$$\text{nABCD}(F_1, F_2) = \frac{W_1(F_1, F_2)}{2 \cdot \text{IQR}_{\text{pooled}}}$$

where $W_1(F, G) = \int |F(x) - G(x)| dx$ is the area between the two CDFs and $\text{IQR}_{\text{pooled}}$ is the IQR of the pooled distribution. The factor of 2 calibrates the scale so that a 1-IQR pure location shift yields exactly $\text{nABCD} = 0.5$.

Properties. Symmetric, scale-free, sensitive to location/scale/shape (unlike SMD, location only); baseline-only — no outcome data needed.

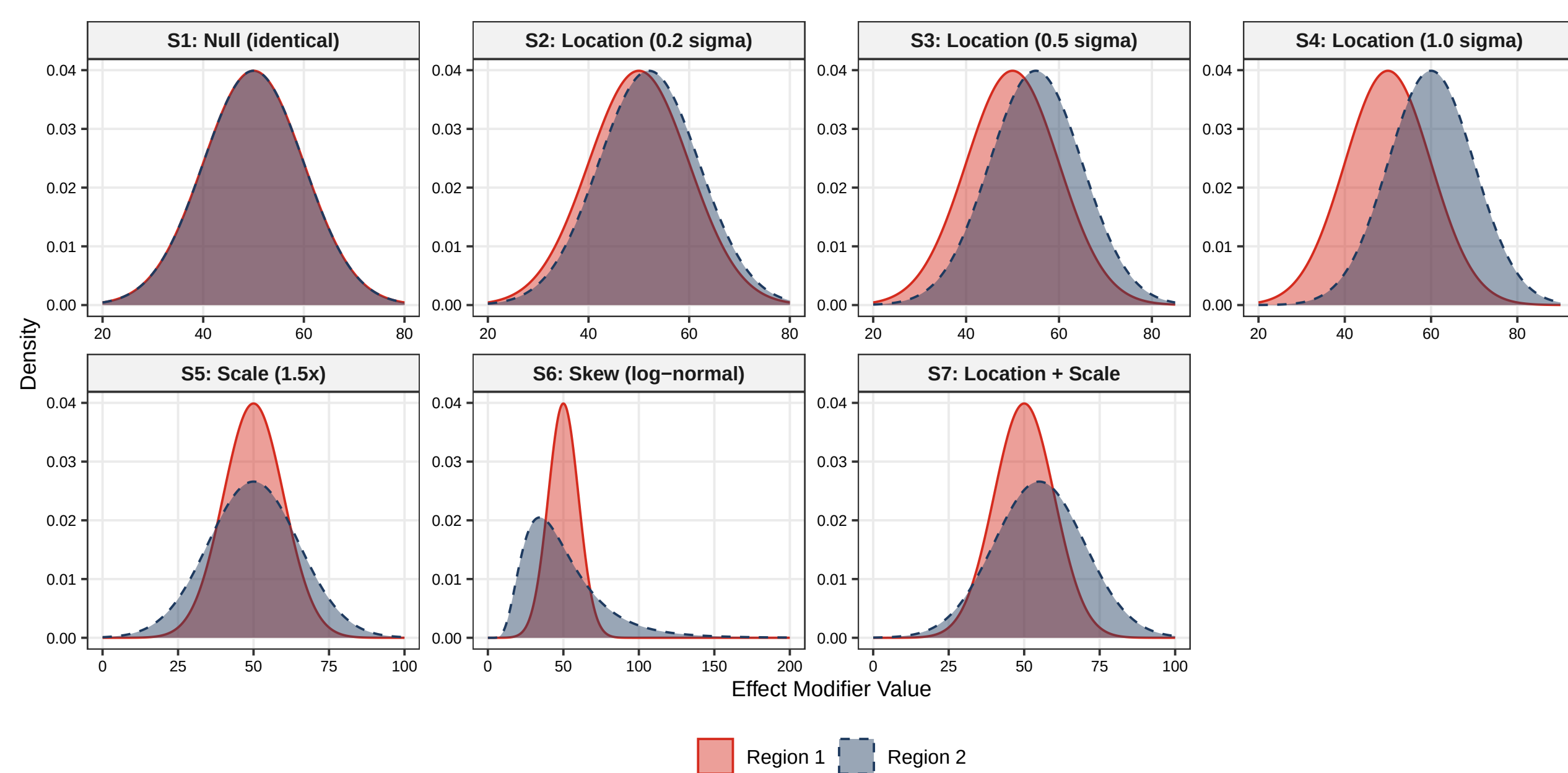


Shaded area between two CDFs equals $W_1(F_1, F_2)$; nABCD divides by $2 \cdot \text{IQR}_{\text{pooled}}$.

Heterogeneity bound. By Kantorovich–Rubinstein duality, if the CATE $\tau(x)$ is L -Lipschitz, the between-region effect difference is bounded: $|\bar{\tau}_1 - \bar{\tau}_2| \leq \blacksquare_{\text{max}} = 2L \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}$. This linkage from distributional metric to clinical scale is specific to W_1 — W_2 and KS admit no analog (Gibbs & Su, 2002).

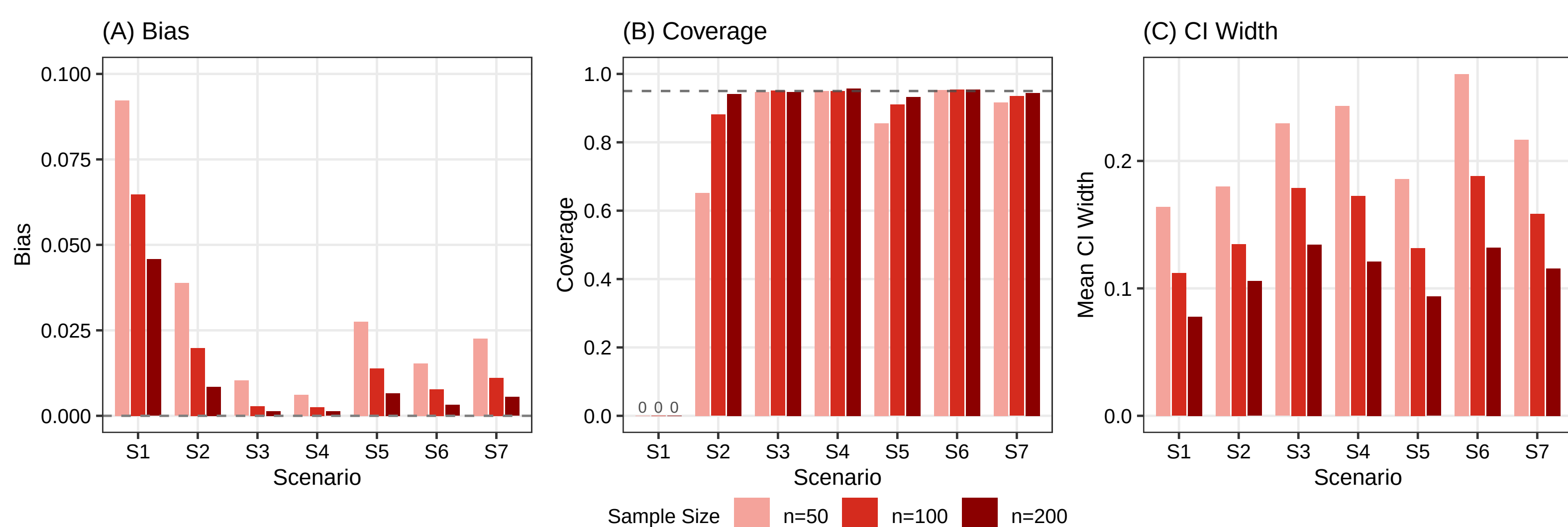
Simulation: Design & Results

Design. Seven scenarios (S1–S7) isolate null, location-shift, scale-shift, skewness, and combined location+scale; 10,000 replications per scenario— n cell, $n \in \{50, 100, 200\}$; bootstrap $B = 2,000$, percentile 95% CIs.



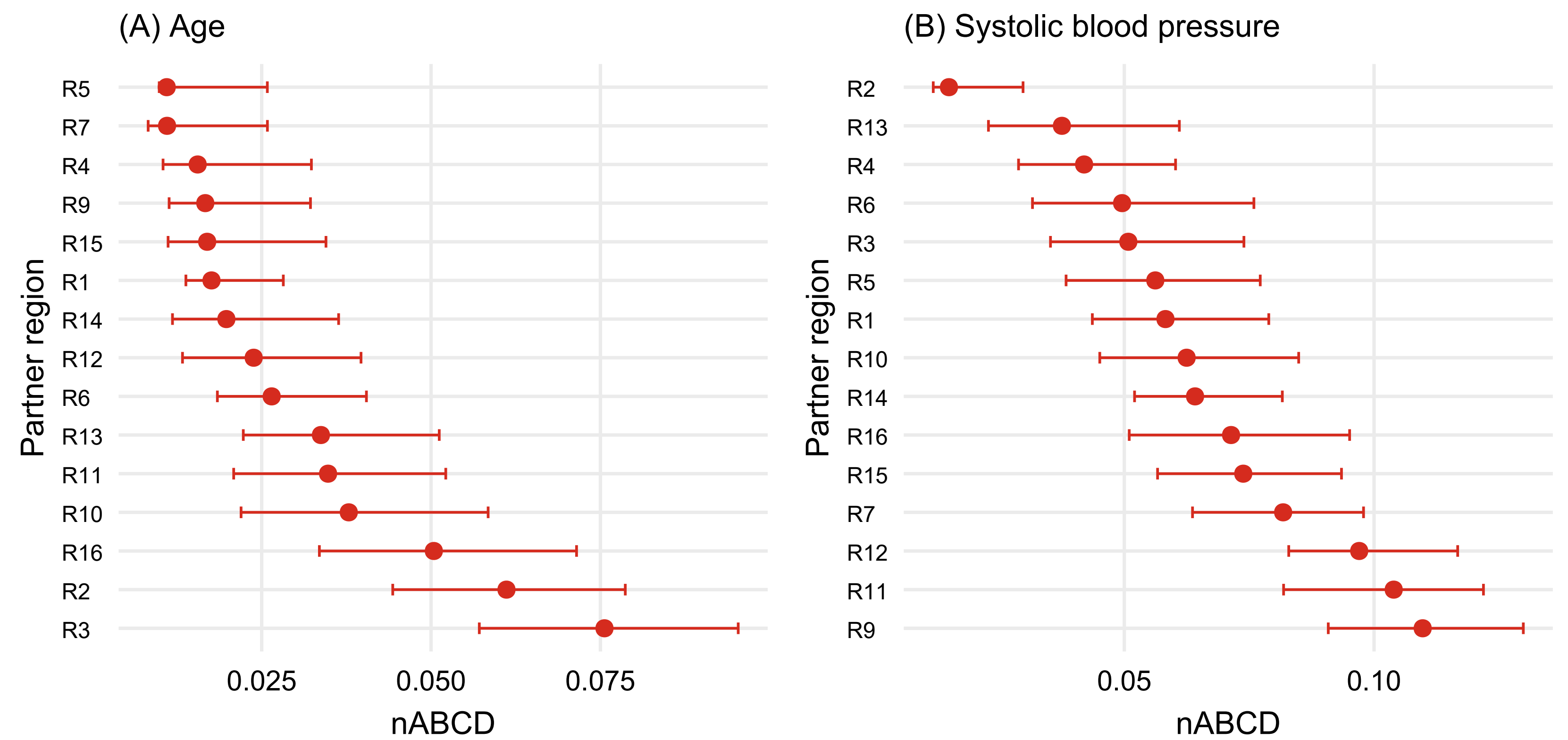
Scenarios S1–S7: Region 1 (red) vs. Region 2 (blue) densities; S1 null, S2–S4 location, S5 scale, S6 skew, S7 location+scale.

Results. Bias < 0.02 and coverage 0.88–0.96 across S2–S7 at $n \geq 100$ (S3, S4, S6, S7 reach nominal 0.95). **Recommendation:** $n \geq 100$ per region; $n = 50$ unreliable near the boundary.



Application: GUSTO-I

Setup. Sponsor develops **Drug T** (thrombolytic for AMI) and plans a Phase 3 MRCT. Planning-stage IPD: **GUSTO-I** ($N = 40,830$; 16 anonymized regions). Anchor: **R8** ($n = 2,916$); 15 candidate partners. Effect modifiers: **age** and **systolic blood pressure**, flagged by Drug T's Phase 2. Class-level CATE slopes unavailable $\Rightarrow L^*$ pathway.



nABCD with 95% percentile bootstrap CIs ($B = 2,000$). Age (A) and systolic blood pressure (B); partners ordered by ascending nABCD .

Joint evaluation is essential. R2 is 2nd-worst on age (0.061) yet best on systolic blood pressure (0.015); R9 is nearly best on age (0.017) yet worst on systolic blood pressure (0.110). Single-modifier rankings give opposite conclusions — candidates must be evaluated jointly.

Eligibility & leading candidate. Plausible CATE-slope upper bounds (FTT 1994): $L_{\text{age}}^{\text{UB}} = 1 \times 10^{-2}/\text{yr}$, $L_{\text{SBP}}^{\text{UB}} = 2 \times 10^{-3}/\text{mmHg}$. A region is eligible if $L^* > L_{\text{clin}}^{\text{UB}}$ at $\blacksquare_{\text{clin}} = 1\%$ pt (30-day mortality). **Six regions are jointly eligible: R1, R4, R5, R6, R14, R15.** **R4** is balanced across both modifiers (3rd-lowest age = 0.016; 4th-lowest systolic blood pressure = 0.042) and is selected as the leading pooling partner for R8.

Clinical Calibration

nABCD is a distance; to translate it into a clinical claim, two pathways are offered depending on what is known about CATE sensitivity.

- **Pathway 1 (L known):** $\blacksquare_{\text{max}}$ is the worst-case regional treatment-effect difference on the clinical scale.
- **Pathway 2 (L unknown):** solve $L^* = \blacksquare_{\text{clin}} / (2 \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD})$; sponsor judges plausibility for the modifier and drug class.

Discussion

Implications

- **Distributional + clinical scale.** The bound $|\bar{\tau}_1 - \bar{\tau}_2| \leq 2L \cdot \text{IQR}_{\text{pooled}} \cdot \text{nABCD}$ is unique to W_1 ; W_2 and KS admit no analog.
- **Beyond SMD.** Captures location, scale, and shape on a clinically interpretable scale; estimation-centered (report $\widehat{\text{nABCD}} + \text{CI} + \blacksquare_{\text{max}}$), not testing.

Strengths

- **Fills the ICH E17 gap** with transparent inputs supporting sponsor judgment, not a fixed cutoff.
- **Joint evaluation** of multiple effect modifiers prevents single-modifier rank reversals (R2 vs R9).

Practical Recommendations

- **Plan with L^*** when class-level CATE slopes are unavailable; report $\blacksquare_{\text{max}}$ when L is known.
- **Sample size $n \geq 100$ per region**; sensitivity over plausible L ; report all candidate modifiers, not best-case.

References

[1] ICH E17 EWG (2017) *Principles for MRCT Design*. [2] Villani C (2009) *Optimal Transport*. Springer. [3] Panaretos & Zemel (2019) *Annu Rev Stat Appl* 6:405. [4] Sommerfeld & Munk (2018) *JRSS-B* 80:219. [5] Gibbs & Su (2002) *Int Stat Rev* 70:419. [6] GUSTO Inv. (1993) *NEJM* 329:673. [7] FTT Group (1994) *Lancet* 343:311.